



Senior Design Project

Sayda: AI Fish Detection System

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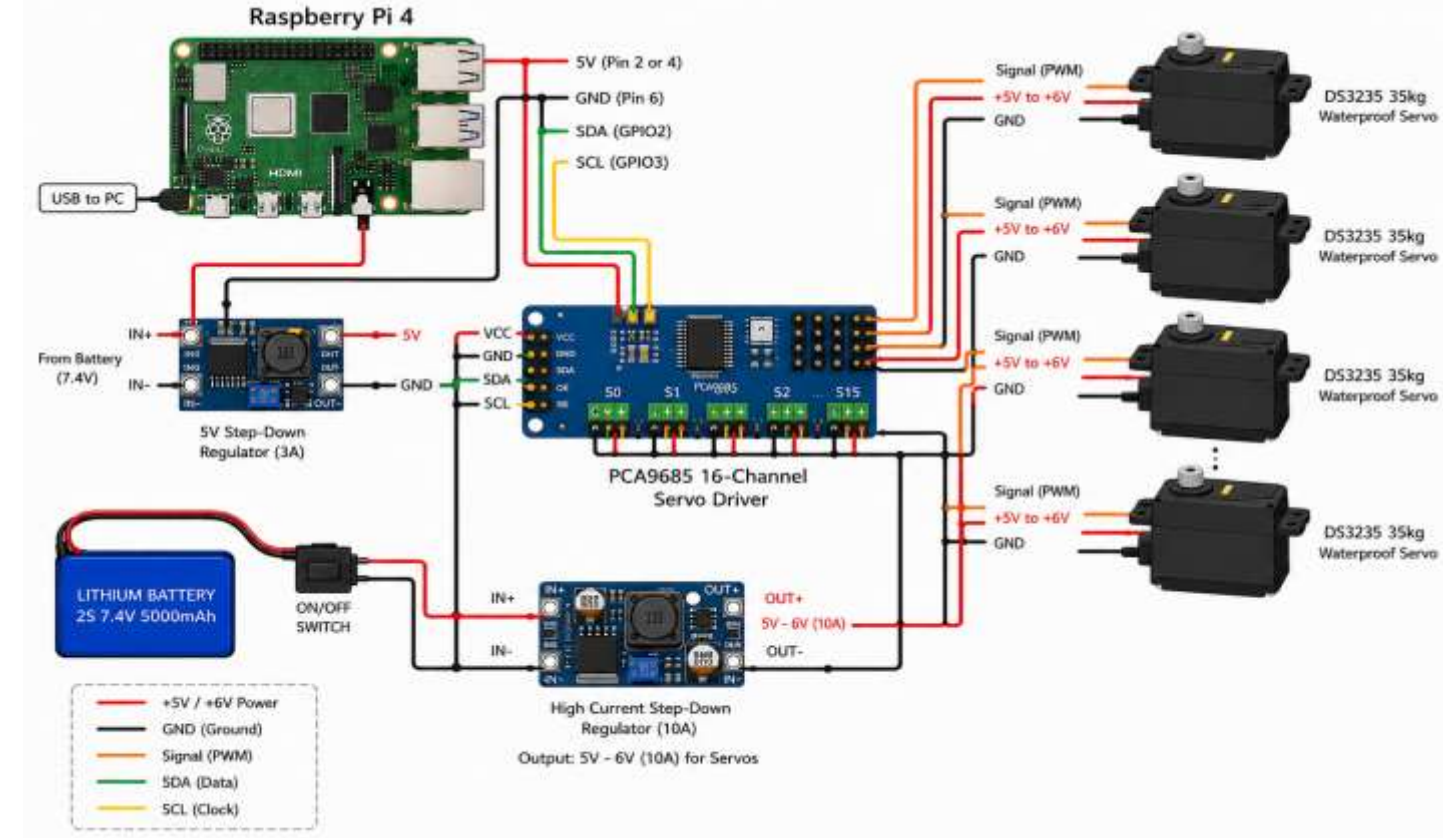
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Abstract

Sayda" is an AI-powered underwater robot designed to eliminate fishing bycatch. Using a Raspberry Pi and YOLOv8, it identifies target marine species in real-time. Upon detection, it instantly triggers a mechanical trap via a smart lock and IP67 servos. Achieving a 4.8ms inference time, Sayda delivers a highly efficient, sustainable solution for selective underwater harvesting.

Methodology

- * *Hardware:* Raspberry Pi, waterproof camera, PCA9685 driver, 4x IP67 servos, smart lock, IP68 enclosure, and a hybrid tether.
- * *Software:* YOLOv8 (7 species) & Python.
- * *Workflow:*
- 1. Capture live video.
- 2. YOLOv8 detects the target.
- 3. Pi triggers servos and lock via I2C



Experimental Investigation

- Testing Setup & Simulation: The system evaluation was strategically divided into two tracks. The AI vision software and graphical user interface were implemented and tested locally using live USB camera feeds.

Concurrently, the mechanical actuation logic—specifically the PCA9685 driver and I2C protocol—was rigorously validated through the Wokwi electronic simulator.

- AI Model Training: The YOLOv8 inference engine was trained over 100 epochs utilizing a custom, balanced dataset sourced from Roboflow.

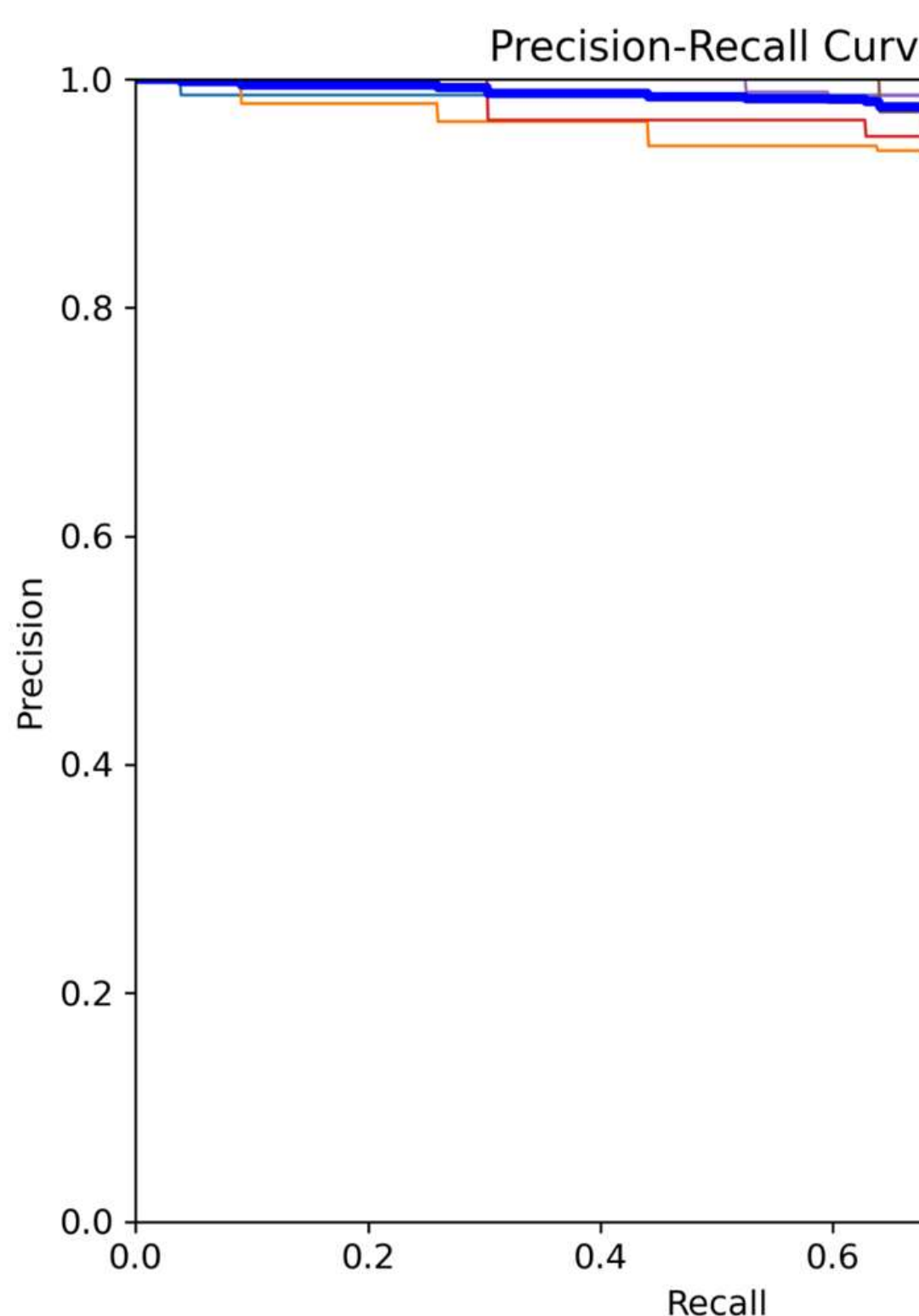
The model was meticulously fine-tuned to accurately classify seven target marine species (Tuna, Agam, Hamour, Hamra, Parrotfish, Salmon, and Tilapia), ensuring steady numerical convergence and preventing overfitting.

- Performance Validation: To guarantee maximum selectivity and completely eliminate false mechanical closures, the model's reliability was systematically tracked.

The evaluation relied on comprehensive metrics, including F1-Confidence curves to find the optimal operational threshold, Precision-Recall (PR) curves, and detailed raw and normalized confusion matrices.

- Latency & Load Stress-Testing: Given the highly dynamic nature of aquatic environments, stringent latency testing was conducted by isolating the computational cycle into pre-processing, core inference, and post-processing stages.

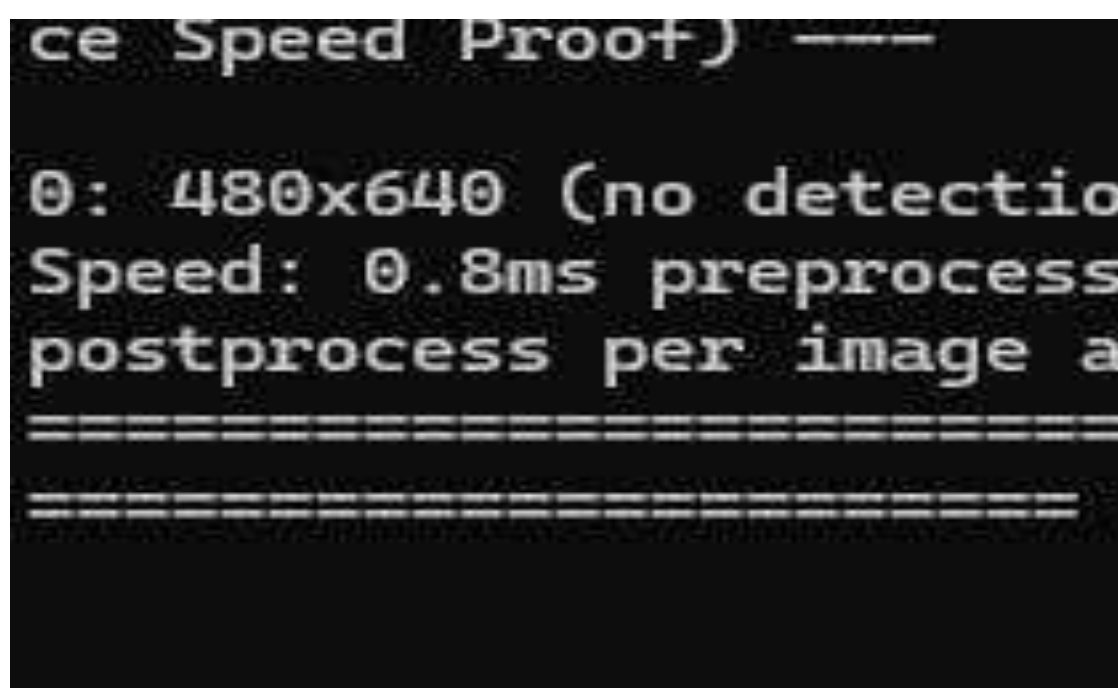
Results



High Accuracy: Achieved near-perfect precision, with near-perfect recall for Hamour (98.8%).

		Confusion Matrix			
Predicted	Tuna	0.91	0.01	0.01	0.07
	Agam	0.89	0.01	0.01	0.09
	Hamour	0.01	0.95	0.01	0.03
	Hamra	0.01	0.01	0.95	0.03
	Parrotfish	0.01	0.01	0.01	0.97
	Salmon	0.01	0.01	0.01	0.97
	Tilapia	0.01	0.01	0.01	0.97
background		0.09	0.09	0.01	0.01
		Tuna	Agam	Hamour	Hamra

Selective Capture:* Confusion matrix showing low inter-species error rates, indicating high selectivity in capturing target species and minimizing bycatch.



Real-Time Speed:* Recorded inference speed of 4.8 ms (total cycle: 6.0 ms) for actuation.

